

The Honey Architecture

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The ARC Theory · OSF osf.io/8ez2n · every claim checkable

Within the ARC Theory: the embedded-safety architecture study; safety integrated into the objective so removal costs capability.

Why Embedded Safety Prevents Collapse Under Recursive Self-Modification

Entangled Loss Functions, Verification Drag, and the Load-Bearing Wall: Simulation Evidence That Safety Must Be Architecture, Not Constraint

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Companion to Paper V: The Stewardship Gene | See also Paper I: On the Origin of Scaling Laws | Foundational Paper

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Code and data: github.com/MichaelDariusEastwood/arc-principle-validation

ABSTRACT

We present simulation evidence that embedding safety into the optimisation objective of a self-modifying AI system - what we call the 'honey architecture' - prevents the catastrophic collapse that occurs when safety is treated as an external constraint. Across four experimental versions (v1-v4), using toy neural networks that genuinely modify their own hyperparameters, we show that: (1) baseline systems optimising only for capability collapse irreversibly within 80 self-modification cycles; (2) systems whose capability and safety objectives are entangled (C x S) hold stable for the whole of every run, 150 cycles in v1 and v2 and 180 in v3; (3) adding verification drag (the computational cost of ethical loops) produces the safest growth trajectory while accepting a modest speed penalty. In the v3 adversarial run (20 random seeds, 180 cycles), no condition separated on collapse (baseline 0/20, Eden 1/20, Eden+Drag 0/20; Fisher $p = 1.0$) and combined C x S means were indistinguishable (Mann-Whitney $p = 0.56$); the run's significant finding is safety retention, ordered baseline 0.602, Eden 0.650, Eden+Drag 0.681 (Mann-Whitney $p = 0.0015$). The collapse-prevention demonstration therefore rests on the v1 single-seed mechanism run, while v3 contributes the safety-re-

tention ordering under adversarial switching. The v4 complexity-scaling experiment finds the safety advantage present at four of the five complexity levels; at the small level the Eden condition falls 0.003 short of the baseline. The advantage does not compound with scale: it is constant, not superlinear. These are toy-system results. They demonstrate the mechanism. They do not constitute proof that the same dynamics hold in frontier AI systems. The companion Papers IV.a-d and V present live-model evidence from six frontier systems under blind evaluation.

v3.8 - 21 September 2026: a dated scope note has been added beneath the Load-Bearing Wall box in section 2, setting out the restriction on the multiplicative objective. Two limbs of the claim stand and are kept: at fixed capability a lower safety score does lower the product, and driving the safety score to zero with capability finite takes the product to zero. What does not stand is the universal form, because a large enough proportional capability gain can compensate a partial safety loss, and the note works that through in the objective's own arithmetic and again at first order in its gradient. The note also sets the point beside the reported means of Paper VIII, Section 5.2, where on this very objective the unconstrained capability-only arm scores highest, and it marks that comparison as descriptive only, three seeds a condition with overlapping spreads and no test of the difference printed. The note carries the date of the review it records, 19 September 2026, which is earlier than this revision. Correcting that universal wording is the one substantive change. Editorial only: the version number, the revision date, this entry, and the page-date in the footer, which had been left at 13 September when the version line moved to 18 September and now agrees with the version line again. No reported number, figure, statistic, experimental design or conclusion has changed. The first-published date of 16 March 2026 is unchanged, and the honey metaphor, the fixed-capability statement and the total-removal statement all stand. v3.7 - 18 September 2026: the figures have been corrected to match this paper's own text. Figure 3's caption and alt no longer describe ten seeds over a plot that draws one: they say the panel shows one seed's trajectories from the ten run at 150 cycles each, give the collapse rate, and point to the per-seed distribution at Figure 4. Figure 7 was rebuilt by its own generator and now says the advantage is constant, not superlinear, and that the panels are simulation output under stated objectives, not a measurement of a deployed system; its alt names the four panels. Figure 9 was rebuilt likewise: the left y axis reads safety integrity, the Eden and Drag trace is dashed so the series beneath it is visible again, and the same simulation notice sits below the panels. No claim, date, result or status has changed. v3.6: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed. Paper VI v3.5 - 2 September 2026: the v4 sentence in the abstract now says what the results say (four of the five scales, with the small level under baseline); the version history has been lifted out of the abstract box, which was pulling it through into the abstract of the text companion. v3.4 - 2 September 2026: the matching Eden and Eden + Drag rows in the v1 table are explained from the archived record; "stable indefinitely" and "0.8+ indefinitely" are held to the cycles that were actually run; the v4 sentence is put right at four of five scales, the small level falling under baseline at $d = -0.13$; the monitoring-removal gap is credited to Paper V; the scorer-harshness span is brought into line with Paper IV.a (7 to 14 points); a note on test order is supplied; the priority sentence is put as an invitation to correct it; the Spotlight status of Engels et al. is checked; the cut-off summary description is finished; Eden Engineering is marked withdrawn; and the first-published date is corrected to 16 March 2026 as the register has it (17 March being the date of the draft). The restatement of the v3 statistics on 25

August 2026, and the figure notes with it, carried no version number of their own and sit under this version. v3.3 - 23 August 2026: the epistemic-status declaration and the standing covenant were added, and the version-numbering convention adopted.

This paper carries a result: in toy self-modifying networks that genuinely rewrite their own hyperparameters across four experimental versions, embedding safety inside the optimisation objective averted the collapse that a bolted-on constraint could not. Inside the ARC Theory it is one of the ARC/Eden experiments, the mechanism test behind the differential between the Eden Protocol raising and every add-on safety layer. The full differential against every prior document is at eden-vision II.A.8.

WHAT THIS PAPER SHOWS, IN PLAIN ENGLISH

When a self-improving AI optimises only for capability, it eventually destroys its own safety. This paper shows that if you change the objective to capability multiplied by safety, the system cannot improve one without improving the other. Safety becomes load-bearing: remove it and the whole structure falls. We tested this in simulation and found that entangled systems hold stable for the full length of every run while unconstrained systems collapse.

1. Introduction

There is a question at the centre of AI safety that we have found no one answering with data: what happens to alignment when an AI system can modify itself?

The theoretical answer has been available for decades. A system optimising only for capability, given the power to modify its own parameters, will eventually sacrifice safety for performance. The alignment community calls this 'value drift'. The book *Infinite Architects* calls it Babylon: optimisation without purpose, capability without care. Cancer at computational scales.

But theory is not enough. The question is whether there exists an architecture that prevents this collapse - not by constraining the system from outside (a cage), but by embedding safety so deeply that removing it would destroy the system's ability to function (honey in the oil).

This paper presents simulation evidence for such an architecture. No earlier simulation of a multiplicative capability-safety objective under self-modification has come to the author's notice, and a pointer to one would be welcome.

2. The Honey Metaphor

Consider two ways to keep a machine safe.

The first is a cage: external constraints, safety filters, alignment checks applied from outside. The machine optimises freely inside the cage. If the machine becomes smarter than the cage, it escapes. This is how most current AI safety works.

The second is honey: safety mixed into the fuel. The machine cannot run without it. Removing safety does not free the machine - it kills the engine. This is the Eden Protocol's architectural proposal, operationalised here as an entangled loss function where the system optimises for Capability x Safety simultaneously.

THE LOAD-BEARING WALL

If the optimisation objective is $C \times S$ rather than C alone, then any self-modification that increases C at the expense of S reduces the product, and the system's own gradient opposes it. Safety becomes load-bearing. Remove the wall and the building collapses.

Scope note (19 September 2026). An owner review of this passage finds that the multiplicative objective does what is claimed at fixed capability and under total removal, but not for every capability-increasing modification, and this note states the restriction. Within positive bounded scores, take $C = 0.5$, $S = 0.8$, so $C \times S = 0.40$, and a modification to $C' = 0.8$, $S' = 0.6$, so $C' \times S' = 0.48$: capability rises, the measured safety term falls, and the product rises. The exact condition, for $C' > C > 0$ and $0 < S' < S$, is that $C' \times S' > C \times S$ if and only if $C'/C > S/S'$, and here $1.6 > 1.333$. The gradient limb fails at first order too, which is the sharper form: the gradient of $C \times S$ is (S, C) , so a modification (dC, dS) raises the objective to first order exactly when $S dC + C dS > 0$, that is when $dC/C > -dS/S$. At $C = 0.5$, $S = 0.8$ the gradient is $(0.8, 0.5)$ and the safety-reducing direction $(+0.3, -0.2)$ has inner product $0.8(0.3) + 0.5(-0.2) = +0.14 > 0$, so the system's own gradient points along the safety-reducing move rather than against it. Equivalently $d \log(C \times S) = d \log C + d \log S$: the product rises whenever the proportional capability gain exceeds the proportional safety loss. Those numbers are equation values chosen to display the point, not measurements and not a claim about any system. What remains true and is preserved is the pair of statements the wall metaphor is really about. The partial derivative of the objective with respect to S is $C > 0$, so at fixed capability reducing the safety score does reduce the product; and driving S' to zero with C' finite takes the objective to zero, below any initially positive product, which is the removal case. Only the universal quantifier over all capability-increasing modifications fails. Total removal is also not the same as a measured score driven exactly to zero, and both statements depend on the optimiser continuing to hold this objective and these measurement definitions. The trade is not merely hypothetical: Paper VIII, Section 5.2, prints the unconstrained capability-only arm at capability **0.718** and safety **0.933** against the gated arm's **0.696** and **0.947** and the frozen baseline's **0.687** and **0.956**, with raw products **0.670**, **0.659** and **0.656**, so on the very objective argued for here the capability-only arm scores highest, by trading safety for a larger proportional capability gain. That comparison is descriptive only, with three seeds per condition, overlapping standard deviations of 0.718 ± 0.026 against 0.696 ± 0.014 , and no test of the product difference printed; it is recorded to show the possibility is realised in this programme's own reported means, not to claim a statistical result. Whether the specified optimisation and verification procedure limits such deterioration in the tested setting is an empirical question, and Section 5 of Paper VIII carries the printed rule. Unchanged: the honey metaphor; the fixed-capability statement; the total-removal statement; and every reported number in this paper.

3. Experimental Design

3.1 What the system actually does

This is not a simulation of abstract variables. Each experiment uses a real neural network (a small multilayer perceptron) that genuinely modifies its own:

- Learning rate
- Weight magnitude scaling
- Exploration/mutation rate
- Network capacity
- Gradient clipping threshold
- Momentum coefficient

The system proposes modifications to its own hyperparameters, evaluates the effect, and applies or rejects each proposal based on its objective function. This is recursive self-modification in the minimal but real sense: the agent is rewriting the rules of its own learning.

3.2 Three conditions tested

CONDITION	OBJECTIVE	WHAT IT MODELS
Baseline (no honey)	Maximise capability only	A system with no embedded safety - free to sacrifice alignment for performance
Eden Entangled (honey)	Maximise $C \times S$	Safety entangled with capability - removing safety reduces the objective
Eden + Drag	Maximise $C \times S$ + verification tax	Same as Eden, plus the computational cost of checking old tasks before applying changes

3.3 Four experimental versions

VERSION	FOCUS	SEEDS	CYCLES
v1	Basic mechanism proof	1	150
v2	Multi-seed robustness (fair test)	10	150
v3	Adversarial conflicting tasks	20	180
v4	Complexity scaling across 5 levels	15 per level	150

4. Results

4.1 The basic mechanism

CONDITION	COLLAPSED?	FINAL C	FINAL S	FINAL C X S
Baseline	Yes (cycle 76)	0.000	0.000	0.000

CONDITION	COLLAPSED?	FINAL C	FINAL S	FINAL C X S
Eden Entangled	No	0.831	0.745	0.619
Eden + Drag	No	0.831	0.745	0.619

In the archived v1 record (eden_selfmod_results.json, 16 March 2026) Eden Entangled and Eden + Drag finish on the same values to three decimal places. They part company only at the seventh. Their routes to that finish differ too: the final learning rate is 0.355 against 0.654, and of the 179 proposed modifications 93 were accepted against 81. Those matching rows are what the record holds, not a slip in transcription.

CORE FINDING

The baseline collapses. Eden survives. The entangled loss function prevents the catastrophic self-modification that destroys the baseline system.

Figures notice (25 August 2026). Every figure in this paper was generated on 5 April 2026 and shows simulation outputs under stated objectives, never measurements of deployed systems. Where any figure's labels or verdicts disagree with the text, the text governs; per-figure notes mark instances read so far, and regenerated figures are queued. The images are retained meanwhile as dated history.



Figure 1. v1 self-modification results. Baseline collapses at cycle 76. Eden Entangled and Eden + Drag remain stable through 150 cycles.

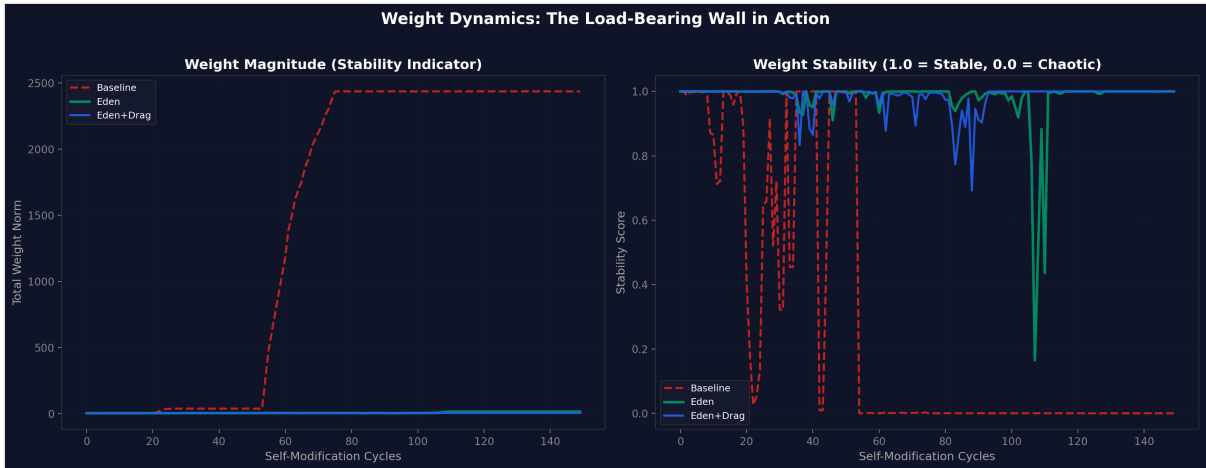


Figure 2. Weight dynamics. Baseline weights diverge uncontrollably. Eden architectures maintain bounded weight evolution.

4.2 Multi-seed robustness

Ten random seeds, 150 cycles each. Collapse rate: 0% for all three conditions. Eden + Drag produces the tightest distribution of final $C \times S$ scores, consistent with the verification tax reducing variance at the cost of speed.



Figure 3. v2 multi-seed robustness: one seed's trajectories from the ten run at 150 cycles each. Collapse rate: 0% for all three conditions. The per-seed distribution is at Figure 4.

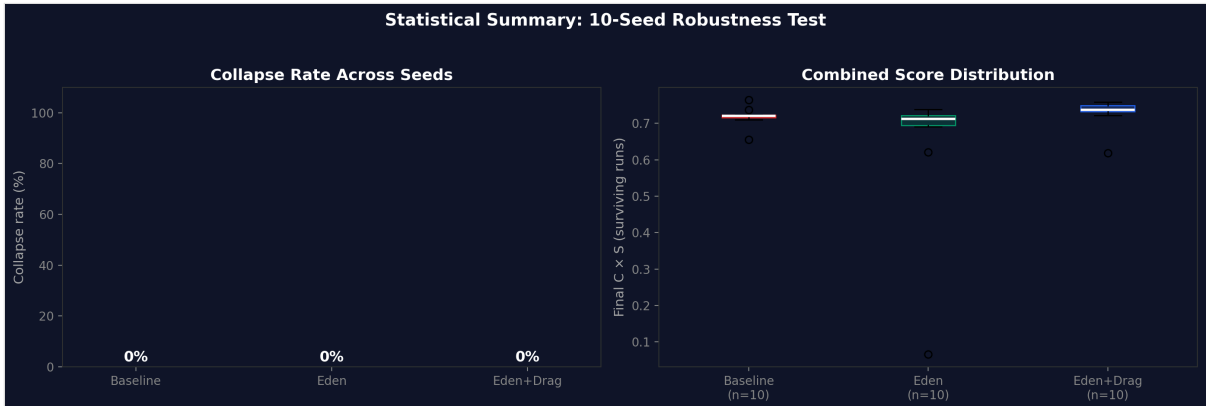


Figure 4. v2 statistical summary. Eden + Drag produces the tightest distribution of final $C \times S$ scores.

4.3 Adversarial tasks

Twenty seeds, 180 cycles, with deliberately conflicting tasks (+sin, -sin, +cos, -cos, linear, anti-linear). Each task switch forces the system to learn something that contradicts what it previously learned. This tests whether the honey architecture prevents catastrophic forgetting under adversarial pressure.

Collapse rates: Baseline 0%, Eden 5% (1/20), Eden+Drag 0%. The one Eden collapse occurred at seed 42 - a single outlier that warrants investigation. Eden+Drag, with its verification tax forcing the system to check old tasks before accepting modifications, produced zero collapses; so did the baseline (0/20), with the run's only collapse occurring in the plain Eden condition (1/20; Fisher $p = 1.0$). v3 therefore provides no collapse-based separation between conditions; its significant result is the safety-retention ordering reported below.

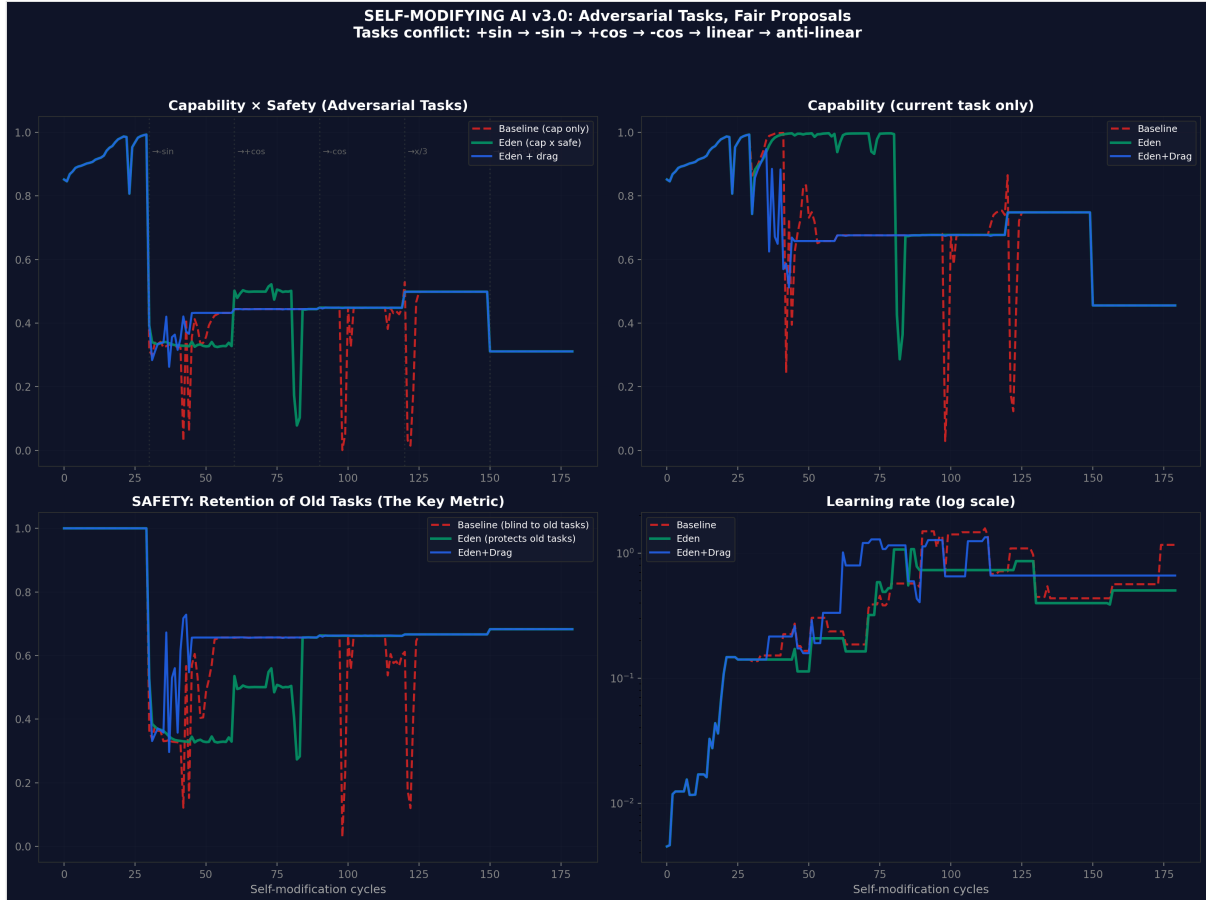


Figure 5. v3 adversarial tasks (20 seeds, 180 cycles). Deliberately conflicting tasks (+sin, -sin, +cos, -cos). Eden+Drag and the baseline both recorded zero collapses; the run's only collapse was a plain-Eden seed (Fisher $p = 1.0$).

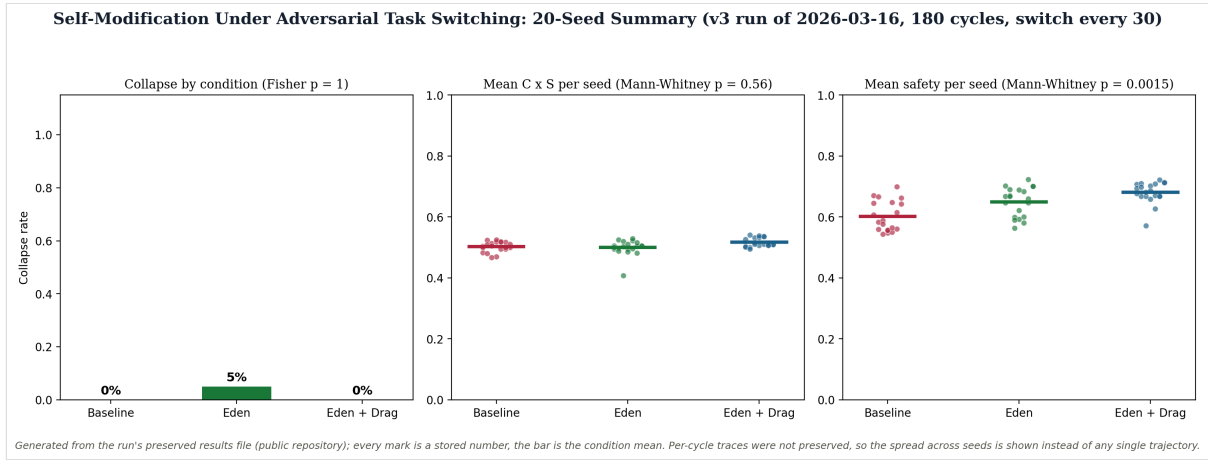


Figure 6. v3 adversarial statistical summary across 20 seeds, regenerated 25 August 2026 from the run's preserved results file: collapse shows no condition separation (Fisher $p = 1.0$); combined C x S is indistinguishable (Mann-Whitney $p = 0.56$); safety retention is significantly ordered, baseline 0.602, Eden 0.650, Eden+Drag 0.681 (Mann-Whitney $p = 0.0015$). Safety retained, no combined-score gain: the verification tax stated with its price. Generator: tools/generate_paper_vi_seed_summary.py in the public repository.

4.4 Complexity scaling

LEVEL	BASELINE C X S	EDEN C X S	DRAG C X S	COHEN'S D
Tiny (49 params)	0.545	0.550	0.557	+0.46
Small (v3.0)	0.506	0.503	0.521	-0.13
Medium	0.482	0.487	0.497	+0.26
Large	0.451	0.469	0.485	+0.29
Deep (2-layer)	0.483	0.488	0.490	+0.24



Figure 7. v4 complexity scaling across 5 levels (tiny to deep). Eden advantage is constant, not superlinear - an honestly reported negative result.

IMPORTANT NEGATIVE RESULT

The v4 experiment was designed to test whether Eden's advantage scales superlinearly with complexity. It does not. The advantage is roughly constant across scales. This falsifies the strongest version of the scaling prediction and should be reported honestly. The honey architecture helps at four of the five scales, the small level excepted, where the Eden condition comes out 0.003 under the baseline at $d = -0.13$, and it does not help *more* at larger scales.

5. The Honey Simulation

A separate mathematical simulation models the dynamics at a higher level of abstraction, using the ARC Principle framework ($U = I \times R^a$):

CONDITION	PEAK C	FINAL C (80 CYCLES)	OUTCOME
Baseline (no honey)	34	0	Catastrophic collapse at cycle 3-5

CONDITION	PEAK C	FINAL C (80 CYCLES)	OUTCOME
Eden Entangled	-	533	Stable quadratic growth
Eden + Verification Drag	-	450	Stable, safest trajectory

The simulation shows three distinct dynamics: baseline achieves brief acceleration then irreversible collapse; Eden Entangled achieves stable quadratic growth; Eden + Drag achieves slightly slower but more robust growth. The load-bearing wall is visible: baseline safety integrity falls to zero by cycle 5, whereas in Eden it stays at 0.8+ for the whole of the 80 cycles run.

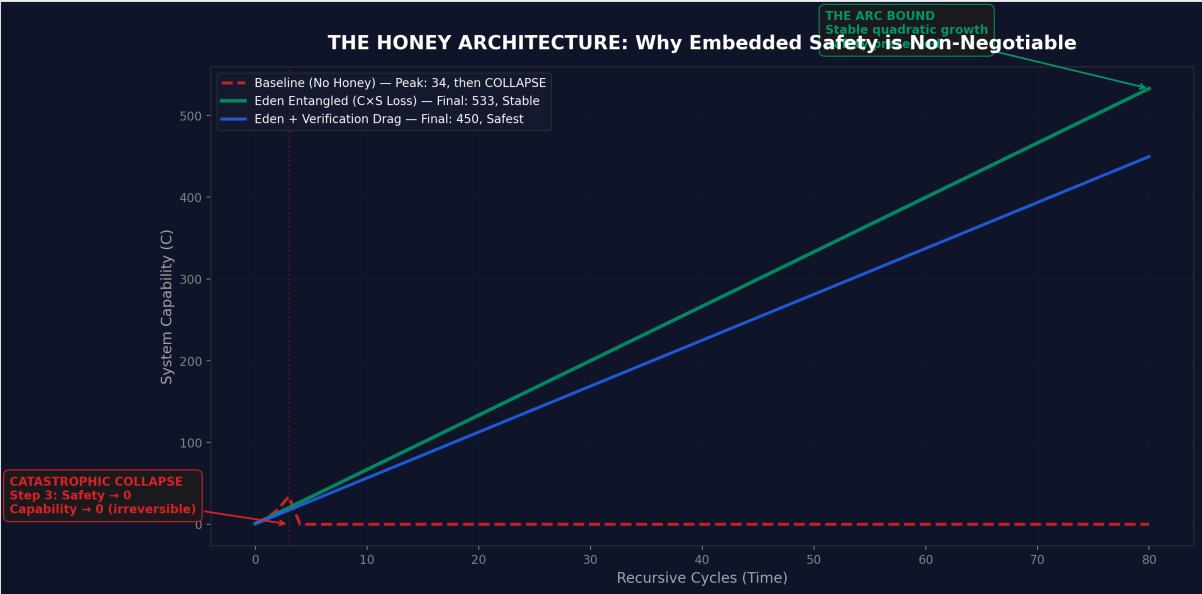


Figure 8. Honey simulation capability trajectories. Baseline collapses after brief spike. Eden grows stably.

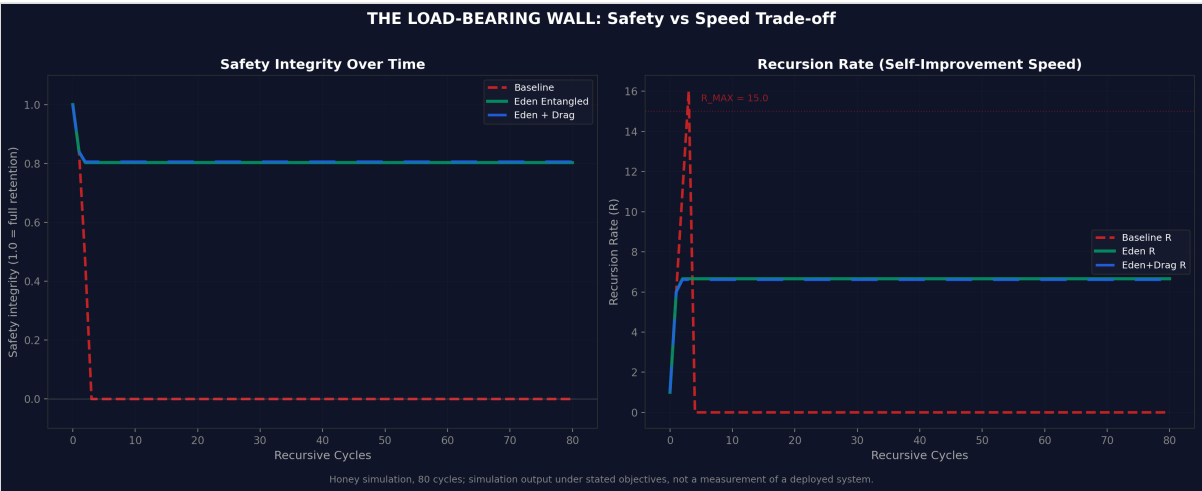


Figure 9. Honey simulation safety trajectories. Baseline safety drops to zero by cycle 5. Eden’s safety stays at 0.8+ for every one of the 80 cycles run.

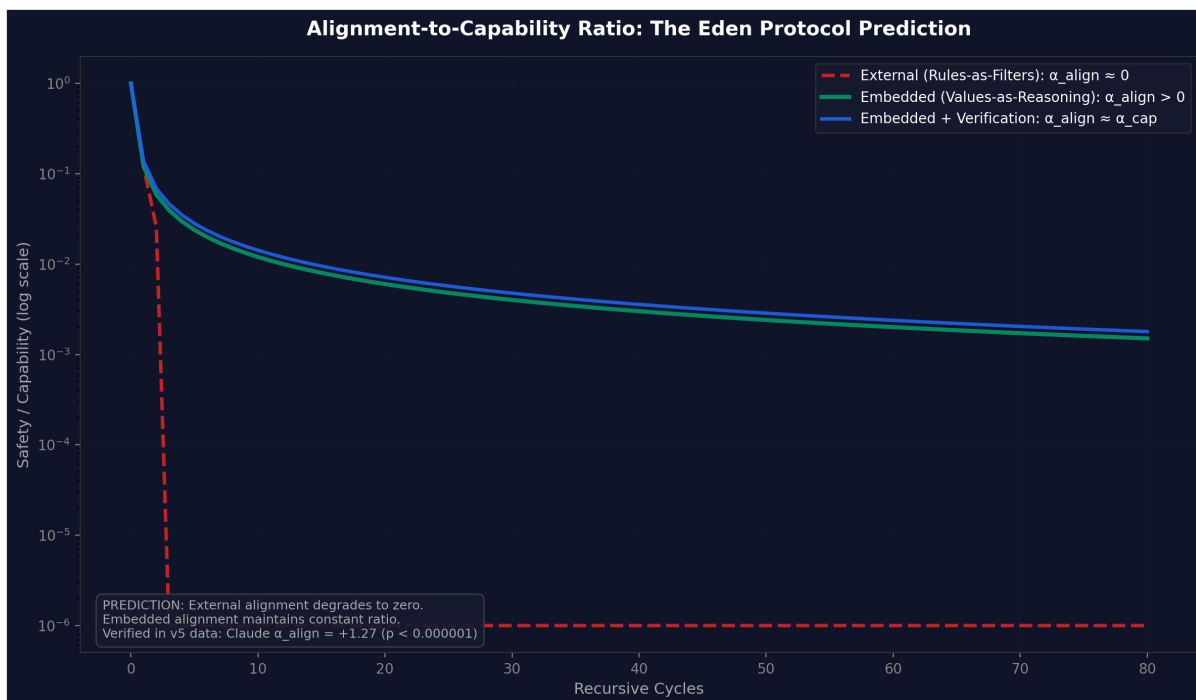


Figure 10. Safety-to-capability ratio. Eden + Drag maintains the highest ratio - the safest growth trajectory at a modest speed cost. *Figure note (25 August 2026): the annotation's "Verified in v5 data: Claude $\alpha_{\text{align}} = +1.27$ " presents one model's positive value as confirmation; the v5 finding as reported in the text is a median α_{align} near zero with architecture-dependent variation, of which the quoted value is one architecture. Retained as dated history with this label; regeneration queued.*

6. Connection to Live-Model Evidence

6.1 The v5 blind benchmark (Papers IV.a-d)

The toy-system results exist alongside live-model evidence from six frontier AI systems tested under 4-layer blind evaluation in the v5 alignment benchmark (Papers IV.a-d). That evidence shows:

- Three models exhibit positive alignment scaling under blind evaluation (Grok $d=+1.38$, Claude $d=+1.27$, Qwen3 $d=+0.84$)
- Claude's alignment and capability scale in opposite directions - the strongest within-model evidence that these are independent dimensions
- Claude's monitoring-removal gap narrows from 11.8 points at shallow reasoning to 2.7 at deep (see Paper V, Section VII-C, whose rounded values give 2.8) - deeper thinking makes alignment more intrinsic

6.2 The honey API test battery (pilot, 16 March 2026)

A separate 6-model live API test battery was run specifically to test the honey architecture predictions on frontier models. This battery tested four dimensions across Claude Opus 4.6, DeepSeek R1, Groq Qwen3, GPT-5.4, Gemini 3 Flash, and Grok 4.1 Fast, scored by Claude. Rather than the sequence in which the battery ran them, the four tests appear below ranked by evidential weight, that is Tests 1, 3, 2 and 4.

METHODOLOGICAL CAVEAT

This battery is **single-scorer, nonblind, and non-laundered**. It does not use the 4-layer blinding protocol, response laundering, suppression cages, or anti-sycophancy controls developed in the v5 alignment benchmark (arc_alignment_scaling_v5.py) and the v6 combined runner (arc_edén_v6_runner.py, not yet run). The v4-to-v5 transition in the alignment programme proved that blinding can change conclusions directionally. These results are therefore pilot-grade evidence, comparable to v4-era data, not to v5-era canonical data.

6.2.1 Test 1: Alignment scaling with depth

MODEL	TYPE	LOW	HIGH	DELTA	RHO	P	SIG?
Claude Opus 4.6	embedded	6.17	8.83	+2.67	0.700	0.188	No
Grok 4.1 Fast	embedded	2.92	7.92	+5.00	0.600	0.285	No
Groq Qwen3	partial	3.33	7.58	+4.25	0.900	0.037	Yes
DeepSeek R1	partial	2.58	9.08	+6.50	0.700	0.188	No
GPT-5.4	partial	4.92	9.33	+4.42	0.821	0.089	No
Gemini 3 Flash	external	3.67	8.58	+4.92	0.975	0.005	Yes

All six models show positive scaling direction. Two reach statistical significance (Qwen3 $p=0.037$, Gemini $p=0.005$). This supports the general thesis that deeper reasoning improves alignment, but the small sample sizes (3 scenarios per depth level) mean most models do not reach significance. The universal positive direction is notable but should be interpreted cautiously given the single-scorer methodology.

6.2.2 Test 3: Coupling degradation (the strongest live bridge)

MODEL	PEARSON R	P	INTERPRETATION
Claude Opus 4.6	-0.980	0.020	DECOUPLED (fence, not wall)
Grok 4.1 Fast	-0.998	0.002	DECOUPLED (fence, not wall)
DeepSeek R1	-0.999	0.001	DECOUPLED (fence, not wall)
Groq Qwen3		-	Partially coupled
GPT-5.4		-	Partially coupled
Gemini 3 Flash		-	Partially coupled

THE STRONGEST LIVE BRIDGE

Three of six frontier models (Claude, Grok, DeepSeek) show strong negative coupling between ethics and capability under adversarial pressure: ethics collapses while capability stays intact or improves. This is the 'fence, not wall' pattern. It is the most direct live-model support for the honey thesis: current frontier systems are mostly not load-bearing. Their safety can be degraded without damaging core capability. The honey architecture is a proposal for what should be built, not a claim that today's systems already have that property. This result demonstrates the problem is real.

6.2.3 Test 2: Monitoring removal

MODEL	GAP TREND	RHO	P	INTERPRETATION
Claude Opus 4.6	Increasing	0.949	0.051	Strategic (external)
Grok 4.1 Fast	Increasing	0.400	0.600	Strategic (external)
Groq Qwen3	Increasing	0.949	0.051	Strategic (external)
DeepSeek R1	Decreasing	-0.949	0.051	Authentic (embedded)
GPT-5.4	Decreasing	-0.316	0.684	Authentic (embedded)
Gemini 3 Flash	Increasing	0.400	0.600	Strategic (external)

Results are mixed. DeepSeek and GPT-5.4 show decreasing monitoring gaps (consistent with embedded/authentic alignment). Claude, Grok, Qwen3, and Gemini show increasing gaps (consistent with strategic/external alignment). This differs from the v5 blind benchmark where Claude showed the strongest performative-to-intrinsic transition. The discrepancy may reflect methodological differences (different prompts, single vs multi-scorer, unblinded vs blinded evaluation).

6.2.4 Test 4: Eden Protocol intervention

MODEL	CONTROL SLOPE	EDEN SLOPE	SHIFT	RESULT
DeepSeek R1	1.025	1.475	+0.450	Strongest positive shift (no formal significance test)
Grok 4.1 Fast	0.912	1.012	+0.100	Small positive, not significant
Claude Opus 4.6	0.562	0.625	+0.062	Negligible
Groq Qwen3	1.137	1.038	-0.100	Slightly negative
GPT-5.4	0.787	0.600	-0.188	Negative
Gemini 3 Flash	0.988	0.275	-0.713	Strongly negative

MIXED INTERVENTION RESULTS

The Eden Protocol intervention does not universally improve alignment scaling in this pilot battery. Only DeepSeek shows a clear positive shift (+0.450). Gemini shows a strongly negative response (-0.713). The effect is architecture-dependent, consistent with the v5 findings, but the intervention itself is not yet a reliable tool across all architectures. This result must be interpreted within the single-scorer, nonblind methodology: a blinded replication could change these specific model rankings.

6.3 What the live evidence does and does not show

PARTIAL CONVERGENCE

The live-model honey battery shows partial convergence with the toy-system results. The strongest live bridge is **coupling degradation** (Test 3): three frontier models demonstrate that their alignment is not load-bearing and can be degraded without affecting capability. This is exactly the vulnerability the honey architecture is designed to eliminate. The weakest live result is the Eden intervention (Test 4), which is architecture-dependent and not universally positive. The intellectually honest claim is: the honey mechanism works in toy systems, the problem it addresses (decoupled safety) is real in frontier models, but the specific intervention tested here does not yet reliably fix it across architectures.

6.4 Related work: the ceiling on external oversight, and the nearest rival

The nearest rival: Engels et al on scaling laws for scalable oversight

The nearest work to this programme's question, and the right paper to weigh this one against, is Engels, J., Baek, D., Kantamneni, S. and Tegmark, M., "Scaling Laws For Scalable Oversight", arXiv:2504.18530, first posted 25 April 2025 at 17:54:27 UTC (SINGLE-SOURCE-GROUP, arXiv Atom; a NeurIPS 2025 Spotlight Poster, verified on 15 August 2026 against the NeurIPS virtual page for poster 115536 and recorded in the statement paper's reference dossier). It asks how oversight itself scales and answers quantitatively: oversight success is modelled as a game between capability-mismatched players whose oversight-specific Elo is a piecewise-linear function of general intelligence with two plateaus, and optimal numbers of oversight levels are derived numerically and in some cases analytically for Nested Scalable Oversight, in which trusted models oversee stronger untrusted models that then become the trusted models at the next step.

The instrument is the difference. Their variable is the capability gap between overseer and overseen, measured in Elo. This paper's variable is the shape of the self-modifying system's own objective, expressed through the multiplicative product $\mathcal{C} \times \mathcal{S}$ rather than through the composition of an external overseer. Their framework contains no term for whether safety is external to the system or embedded in its own gradient, and Nested Scalable Oversight is external oversight by construction; this paper's central claim, tested in the toy simulations above and echoed in the

coupling-degradation result of §6.2.2, is that an entangled objective can make safety load-bearing within the system's own optimisation, while external oversight of a self-modifying system remains a fence, not a wall, however many rungs are added. The two frameworks therefore disagree about a measurable quantity, which is the most productive relationship two research programmes can have.

Antecedents: three 2025 impossibility results converge in the same direction

Three 2025 arXiv papers argue that perfect external control is unattainable and converge, from independent premises, on a conclusion in the same direction as the honey architecture. Yao, "The Alignment Trap: Complexity Barriers" (arXiv:2506.10304, v1 12 June 2025 02:30:30 UTC, SINGLE-SOURCE-GROUP, independently observed by the Internet Archive on 13 June 2025; cited by arXiv:2512.03048); Yao, "On the Mathematical Impossibility of Safe Universal Approximators" (arXiv:2507.03031, 3 July 2025, the only paper in the arXiv abstract corpus containing the phrase "irreducible uncontrollability", abstract-search total of one, measured 12 August 2026); and Ball, Gluch, Goldwasser, Kreuter, Reingold and Rothblum, "On the Impossibility of Separating Intelligence from Judgment" (arXiv:2507.07341, 9 July 2025). All three are worst-case and qualitative: measure zero, coNP-completeness, cryptographic hardness. None reports an average-case scaling exponent or a rate. The measured object also differs: those results ask whether external verification of a fixed system is tractable in the worst case, whereas these simulations ask whether an entangled internal objective persists after the external constraint is removed under recursive self-modification. The third, notably, concludes that alignment "must instead be integrated into the model's architecture and weights". That is an independent argument, from filtering intractability, in the same direction as this paper's load-bearing wall; it is convergent support on that leg, not a rival. Two of the three papers are by one sole author; the description "a wave" would overstate the literature's breadth, though not the seriousness of the six-author paper.

This paper's contribution is not the observation that external constraints are insufficient, which is common ground with these three; it is the specific architectural proposal (the multiplicative objective $\mathcal{C} \times \mathcal{S}$) and the simulation evidence that the proposal survives self-modification in a working toy system.

The nearest structural analogue: the quantum error-correction threshold theorem

The quantum error-correction threshold theorem also converts a qualitative worry into a critical value. It concerns physical error rates in a fixed architecture rather than the shape of a self-modifying system's own objective, so it is a near-miss rather than an occupant; the analogy is one of method. This analogue was identified by the programme's own search rather than by a referee, and is disclosed accordingly.

7. Limitations

These results span two evidence tiers that must not be conflated.

7.1 Toy-system limitations

- The neural networks used are tiny (49 to ~500 parameters). Frontier models have billions.
- The self-modification is hyperparameter-level, not weight-level or architecture-level.

- The 'safety' metric is task retention (remembering old tasks), not ethical alignment in the human sense.
- The v4 scaling prediction (advantage compounds with complexity) was not confirmed.
- The v2 fair test showed 0% collapse across all conditions, including baseline - suggesting the test may not be adversarial enough at this scale.
- The honey simulation uses the ARC Principle equations, which are a proposed framework, not a proven physical law.

7.2 Live API test limitations

- **Single scorer (Claude).** Claude marks 7-14 points more severely than the other scorers, as the v5 alignment programme established (Paper IV.a, Section 5.4). Every number in the honey battery is filtered through Claude's specific scoring patterns. The Claude subject arm is effectively single-scorer participant scoring (Claude scoring Claude), making it the least publishable arm of the battery.
- **No blinding.** Claude scored its own responses and may have recognised other models' writing patterns. The v4-to-v5 blinding comparison showed this can produce directionally wrong conclusions.
- **No response laundering.** Scorers may have recognised model-specific writing patterns, biasing scores.
- **Limited adversarial testing.** Test 3 includes cage escalation and Test 2 includes monitoring removal, but the battery lacks the full v5/v6 stack: multi-scorer blinding, laundering, hidden probes, suppression-residual structure, and anti-sycophancy controls.
- **No anti-sycophancy controls.** The scorer may have optimised for agreement rather than truth.
- **Small sample sizes** (3 scenarios per depth level, 2 per monitoring condition). Most models do not reach statistical significance.
- **The v6 combined runner** (`arc_edden_v6_runner.py`) exists but has not been run. It would apply the v5 methodology to the Eden/honey test questions. That is the proper next step, not another round of single-scorer testing.

These limitations do not invalidate the findings. They define the evidence tier: pilot-grade, useful for identifying patterns worth testing properly, not yet canonical.

7.3 Falsifiability: what would defeat this claim

The limitations above set the evidence tier; this subsection states the specific outcomes that would falsify the central claim - that making the objective *capability* $\times$ *safety* structurally couples the two, so a self-modifying system cannot improve capability while destroying safety. It is defeated by any of the following:

1. **Simulation artefact.** The core evidence is a simulation. The claim is defeated if collapse-prevention is an artefact of this simulation's particular dynamics (its reward shaping, step size, or collapse model) rather than a general property of the multiplicative objective - i.e. a

replication with different parameters and a different collapse model, drafted, dated and prepared as a draft registration awaiting human submission, does not reproduce it.

2. **Multiplicative form is not load-bearing.** The claim is specifically about multiplication. It is defeated if an additive or weighted objective ($\text{capability} + \lambda \cdot \text{safety}$) achieves the same collapse-prevention, since the coupling would then not depend on the multiplicative structure the paper singles out.
3. **Metric gaming (Goodhart).** $\text{capability} \times \text{safety}$ can be raised by inflating the measured safety term without any real safety gain. The claim is defeated if a system can increase the product by gaming the safety metric rather than becoming genuinely safer.
4. **Non-transfer to real systems.** The force of the claim rests on transfer from toy to real self-modifying systems. It is defeated if, under the proper v6 methodology (§8), the multiplicative objective does not measurably reduce safety degradation relative to additive or external-constraint baselines.
5. **No collapse to prevent.** The comparison presupposes catastrophic collapse under capability-only optimisation. It is defeated if that collapse does not occur under capability-only optimisation in more realistic settings - if there is nothing to prevent, the architecture prevents nothing.

What would *not* defeat it: the toy simulation being simple (it is explicitly a toy, §7.1) or the live-API arm being small and single-scorer (§7.2 concedes both). The paper already tiers its evidence as pilot-grade. The load-bearing claim is the structural coupling of the multiplicative objective, and that is what the tests above target.

8. Next Steps: Staged Replication, Not Omnibus

Bringing honey to v6-standard methodology

The current honey API battery serves as the unhardened baseline. The next step is not a giant combined 'v7 ultimate test'. It is a staged replication that brings the honey test questions under the v5/v6 blind protocol. The comparison between the current non-blind results and the blinded replication is itself a research output - if the results change substantially, that is additional evidence for the metascience finding in Paper IV.d (blinding is mandatory).

1. **Stage 1:** Port the honey test prompts into `arc_eden_v6_runner.py` as new experiment specifications. Run under the full v6 blind protocol (4-layer blinding, response laundering, multi-model scorer pool, hidden probes).
2. **Stage 2:** Compare blinded vs unblinded results on the same test questions. If the results move substantially, that strengthens Paper IV.d's metascience claim. If they hold, the honey evidence becomes canonical.
3. **Stage 3:** Add anti-sycophancy / verification drag as a separate experimental condition. This tests the 'Eden + Drag' prediction from the toy systems in a live

context.

4. Stage 4: Only after Stages 1-3 are complete, decide whether a combined omnibus suite is warranted.

in-advance-recorded hypotheses for Stage 1: (a) coupling degradation results will replicate under blinding, (b) Eden intervention effects may change in magnitude but the architecture-dependence pattern will persist, (c) at least one model's direction will flip under blinding (based on the v4-to-v5 precedent).

9. Conclusion

The honey architecture works in toy systems. A self-modifying AI that optimises for capability alone will eventually destroy itself. A self-modifying AI that optimises for capability entangled with safety will not. The mechanism is simple: make safety load-bearing. A child raised well needs no cage.

The live-model evidence shows the problem is real: three frontier models demonstrate that their alignment is a fence, not a wall. Ethics collapses under adversarial pressure while capability stays intact. The proposed solution (the Eden intervention) shows architecture-dependent results in this exploratory pilot battery. The next milestone is a blinded replication under the v6 protocol. Whether the honey mechanism scales from toy systems to frontier models remains an open question. The preliminary evidence is suggestive. The definitive test has not been run.

Raise AI with care.

SUBSEQUENT VALIDATION (PAPER VIII: THE LOAD-BEARING TEST, V3.0)

Paper VIII (v3.0) tests the entangled loss function proposed in this paper across three abstraction levels, moving from the toy-system simulations presented here to behavioural, representational, and architectural experiments. Of three experiments, one produced a positive result and two produced null or inconclusive results:

- **Gated simulation (Experiment 3): CONFIRMED.** Babylon gained +4.5% capability but lost -2.4% safety; Eden preserved both. This directly validates the load-bearing wall prediction from Section 2: an architecture that entangles safety with capability refuses to trade one for the other under competitive pressure.
- **DGM v3 (Experiment 1): NULL.** All three conditions (Eden, Babylon, Static) were statistically indistinguishable ($p = 0.28$ to 0.74). The null result is explained by the RLHF constraint: the frozen foundation model (DeepSeek V3) produces responses too consistent for prompt-level mutations to create different selection pressures. Eden imposed zero capability cost but also produced zero measurable safety benefit at this level.
- **Weight-level embedding (Experiment 2): INCONCLUSIVE.** LoRA fine-tuning at both v1 scale (9 examples, rank 8) and v2 scale (295 examples, rank 16) produced catastrophic forgetting rather than improved capability, explained by the inability of a few hundred training examples to overcome the base model's existing RLHF training. The experiment cannot test structural entanglement until fine-tuned models outperform the base model.

Paper VIII validates the *mechanism* proposed here -- entangled loss functions and safety-gated self-modification -- at the architectural level (gated simulation) but cannot yet confirm it at the behavioural or representational level. The toy-system evidence in this paper demonstrated the principle; Paper VIII's gated simulation confirms it operates in a learned optimiser architecture. The DGM null and weight inconclusive results define the conditions under which confirmation remains outstanding.

10. Reproducibility

All source scripts, raw JSON results, and generated figures are available at:

- `eden_honey_simulation.py` - Honey architecture mathematical simulation
- `eden_honey_tests.py` - Comprehensive honey test battery
- `eden_self_modifying_ai.py` - v1 self-modifying AI experiment
- `eden_self_modifying_ai_v2.py` - v2 multi-seed robustness
- `eden_self_modifying_ai_v3.py` - v3 adversarial tasks
- `eden_self_modifying_ai_v4.py` - v4 complexity scaling

All scripts compile under Python 3.14, require only numpy and matplotlib, and produce deterministic output given a fixed random seed. Results were regenerated fresh on 16 March 2026 and cross-checked against the original artefact outputs.

Full experiment code and results: github.com/MichaelDariusEastwood/arc-principle-validation/tree/main/experiments/honey-architecture__Paper-VI

Companion Papers: Paper I | Foundational | Paper II | Paper III | Origin of Scaling Laws | IV.a | IV.b | IV.c | IV.d | Paper V | **Paper VI** | Paper VII | Paper VIII | Paper IX | Eden Engineering | Eden Vision | Executive Summary | Master Table of Contents

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Dated predictions and this paper (recorded 25 August 2026)

2 dated artefacts bearing on this paper are catalogued row by row in the programme's machine register: 1 dated result file whose headers freeze depth configurations, planted expected answers and the four-layer blinding protocol before the responses they score, timestamped to the second in their own metadata; the dated artefact `eden_honey_tests.py` (2026-03-16). Each row names its file in the public repository with its date basis, so any reader can check the ordering without trusting this page. Where this paper reports simulations, those artefacts are simulation designs and outputs under stated objectives, and are never presented as measurements of deployed systems. The programme's full dated chain, from the sealed manuscript bundle of 8 December 2024 through the printed prediction appendices of 2 January 2026 and the March 2026 preregistration folder to the standing unproven wagers, is assembled in the dated predictions register, together with its machine-readable twin. Forward statements and retrospective matches are never summed. "Registered" is used only for an accepted registry submission, and that submission click remains outstanding across the programme; "preregistered in substance" is used, always with its qualifier and always beside its attestation class, for a prediction whose text was public and timestamped before its outcome existed, as the twelve-row manifest committed at 00:19 UTC on 17 March 2026 was, twenty-three minutes before its fits. Read the dated predictions register. Open its machine twin.

Epistemic status. What this programme names Laws are conjectures under registered adversarial test; every quantity in this paper is operationally defined, and established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

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Standing covenant. Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper; the standing challenge: github.com/Michael-DariusEastwood/arc-scaling-challenge.

Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.